

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JnanaSangama,Belgaum-590018



Mini Project Report on

“AUROMATED VACCINE DELIVERY SYSTEM”

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards the completion of the **Mini Project** under the **Interdisciplinary Project-Based Learning Laboratory**,
Department of Basic Sciences.

by

SN	Name	USN/Roll number
01.	Swayansh Siddh Supkar	1CR25CS195
02.	Syed Aman Alam	1CR25CS196
03.	S Manasa	1CR25EC176
04.	S Gowrish	1CR25EC175

Under the Guidance of
Prof. Naveen kumar
Asst. Professor, Dept. of ECE



CMR INSTITUTE OF TECHNOLOGY

#132, AECS LAYOUT, IT PARK ROAD, KUNDALAHALLI,
BANGALORE-560037

CMR INSTITUTE OF TECHNOLOGY

#132,AECSLAYOUT,ITPARKROAD, KUNDALAHALLI,

BANGALORE-560037

DEPARTMENT OF BASIC SCIENCES



CERTIFICATE

This is to certify that the File Structures mini project entitled “AUTOMATED VACCINE DELIVERY SYSTEM” has been successfully carried out by Swayansh Siddh Supkar(K-49), Syed Aman Alam (K-50), S Manasa (O-50) and S Gowrish (O-49), Bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the Second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Interdisciplinary Project-Based Learning Laboratory, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

Signature of Guide

Prof. Naveen kumar

Dept. of ECE, CMRIT

Signature of HOD

Dr. Raveesha K H
HOD

Dept. of Physics, CMRIT

External Viva

Name of the examiners

Signature with date

1.

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Swayansh Siddh Supkar (K-49)

Syed Aman Alam (K-50)

S Manasa (O-50)

S Gowrish (O-49)

Abstract

The Automated Vaccine Delivery System is an embedded solution designed to ensure safe and efficient transportation of vaccines with real-time monitoring and minimum human intervention. The main objective of this project is to maintain the required temperature conditions for vaccines during transportation, monitor payload status, and provide live location tracking for effective delivery management. The system uses an **Arduino Uno (ATmega328P)** microcontroller interfaced with a **DS18B20 digital temperature sensor** to continuously monitor storage temperature, a **NEO-6M GPS module** for real-time location tracking. Status indications are provided through **LED indicators**, while a **piezo buzzer module** alerts users in case of abnormal temperature conditions or delivery issues. The system is powered using a **12V DC power supply with a 5V voltage regulator** to ensure stable and reliable operation. This project provides a cost-effective and dependable solution for vaccine transportation and monitoring in healthcare applications.

Keywords:

- Vaccine delivery system
- Real-time temperature monitoring
- GPS tracking system
- Arduino-based automation
- Cold chain management
- Healthcare transportation
- Vaccine safety monitoring

CHAPTER 1

INTRODUCTION

- An Automated Vaccine Delivery System is a modern technological solution designed to ensure the safe transportation and monitoring of vaccines during delivery. The system continuously monitors critical parameters such as storage temperature, payload weight, and delivery location, which are essential for maintaining vaccine quality and effectiveness. Sensors and tracking modules are connected to a microcontroller that collects and processes data in real time. In traditional vaccine transportation, monitoring is done manually, which may lead to temperature fluctuations, improper handling, delayed deliveries, and vaccine wastage. Maintaining the vaccine cold chain is one of the major challenges in healthcare logistics. The automated vaccine delivery system overcomes these limitations by providing real-time monitoring, automated alerts, and accurate tracking throughout the transportation process. The system uses an Arduino-based embedded platform integrated with temperature sensors, GPS tracking modules, LED indicators, and buzzer alerts.
- These components help ensure that vaccines are stored under safe conditions and delivered efficiently. The system can also provide location updates and warning notifications when abnormal conditions occur. By reducing human effort and improving monitoring accuracy, the automated vaccine delivery system contributes significantly to reliable healthcare transportation and vaccine safety

1.1 BRIEF HISTORY OF AUTOMATED VACCINE DELIVERY SYSTEM

- The history of vaccine delivery systems is closely connected to the development of healthcare logistics and cold chain technology. Initially, vaccines were transported manually using insulated containers with ice packs, and temperature monitoring relied entirely on human supervision. These traditional methods were time-consuming and often unreliable, resulting in vaccine spoilage due to improper storage conditions.
- During the late 20th century, electronic temperature monitoring devices were introduced to improve vaccine storage and transportation safety. Basic refrigeration systems and digital thermometers helped maintain controlled temperatures during delivery. However, these systems still lacked real-time monitoring and tracking capabilities.

Project title: AUTOMATED VACCINE DELIVERY SYSTEM

- With advancements in embedded systems and IoT technology in the 2000s and 2010s, modern vaccine delivery systems began incorporating GPS tracking, wireless communication, and automated monitoring features. These systems enabled healthcare providers to track vaccine transportation in real time and receive alerts whenever storage conditions became unsafe.
- Today, automated vaccine delivery systems use smart sensors, cloud-based monitoring, and intelligent analytics to improve vaccine transportation efficiency, reduce wastage, and support large-scale immunization programs. This evolution has transformed vaccine logistics into a safer, more reliable, and technology-driven process.

1.2 MODERN AUTOMATED VACCINE DELIVERY SYSTEM

- A modern automated vaccine delivery system uses advanced sensors and embedded technologies to ensure safe and efficient vaccine transportation. It continuously measures important parameters such as storage temperature, payload weight, and delivery location to maintain proper vaccine conditions.
- The collected data is processed using a microcontroller such as Arduino Uno and can be monitored in real time. GPS technology allows live location tracking of the vaccine carrier, while temperature sensors ensure that vaccines remain within the recommended storage range. confirm proper handling during delivery.
- LED indicators and buzzer alerts provide warning notifications whenever abnormal conditions such as temperature changes or delivery issues occur. The entire system is powered using a regulated power supply for stable operation. These systems are widely used in hospitals, healthcare centers, medical logistics, and vaccination campaigns to improve delivery reliability and reduce vaccine wastage.

CHAPTER 2

PROBLEM STATEMENT

2.1 DESCRIPTION

- Traditional vaccine transportation methods rely heavily on manual supervision and periodic monitoring, which can lead to vaccine spoilage, delayed deliveries, and improper handling. Maintaining the required storage temperature throughout transportation is a major challenge, especially in remote areas and long-distance deliveries.
- In many cases, healthcare workers cannot continuously monitor vaccine conditions or accurately track the delivery vehicle in real time. Sudden temperature fluctuations, handling errors, or delays may reduce vaccine effectiveness and increase wastage. Manual monitoring also requires significant human effort and may not provide immediate alerts during emergencies.
- Therefore, there is a need for an automated, reliable, and cost-effective system that can continuously monitor vaccine transportation conditions, provide real-time location tracking, and generate alerts whenever unsafe conditions occur. The automated vaccine delivery system addresses these challenges by integrating sensors, GPS technology, and embedded systems to improve vaccine safety and transportation efficiency.

2.2 CHALLENGE STATEMENT

- Before using an automated vaccine delivery system, vaccine transportation depends mainly on manual checking and traditional cold storage methods. Healthcare workers must physically inspect temperature conditions and delivery status, which is time-consuming and often inaccurate.
- There is no continuous real-time monitoring of vaccine temperature or transportation conditions. Delays in delivery or unexpected temperature variations can damage vaccines and reduce their effectiveness. Lack of proper tracking also makes it difficult to monitor delivery routes and ensure timely transportation. Managing vaccine transportation manually increases operational costs, human effort, and the possibility of vaccine wastage.

CHAPTER 3

3.1 DESIGN THINKING PROCESS

- a) Empathize: Interviews with healthcare workers, delivery personnel, and medical staff revealed problems such as temperature fluctuations, delivery delays, lack of tracking, and vaccine wastage during transportation.
- b) Define: The main requirements identified were accurate temperature monitoring, reliable GPS tracking, automated alert systems, and stable power management.
- c) Ideate: Several ideas involving IoT tracking systems, automated alert mechanisms, and sensor-based monitoring were generated. The final solution selected was an Arduino-based automated vaccine delivery monitoring system.
- d) Prototype: A hardware prototype consisting of Arduino Uno, DS18B20 temperature sensor, NEO-6M GPS module, LEDs, and buzzer was developed.
- e) Test: Testing was conducted under different environmental conditions. The system successfully monitored temperature, tracked delivery location, and generated alerts during abnormal conditions.

3.2 METHODOLOGY

- The methodology of the automated vaccine delivery system focuses on continuous monitoring, tracking, and alert generation during vaccine transportation. Initially, sensors such as the DS18B20 temperature sensor and load cell are installed inside the vaccine storage container.
- The sensor data is processed using an Arduino Uno (ATmega328P) microcontroller, which acts as the central control unit. The temperature readings are continuously compared with predefined safe storage limits required for vaccine preservation.
- The NEO-6M GPS module provides real-time location tracking of the delivery vehicle or vaccine carrier.
- If the temperature exceeds the safe range or any abnormal condition occurs, the buzzer and LED indicators are activated to alert the user. The regulated power supply ensures stable system operation throughout transportation.

Project title: AUTOMATED VACCINE DELIVERY SYSTEM

3.2 WORKING PROCEDURE AND FLOWCHART DESCRIPTION

The automated vaccine delivery system continuously monitors vaccine storage conditions and transportation status. The DS18B20 sensor measures storage temperature. The GPS module continuously tracks the delivery location.

The Arduino Uno processes all sensor data and compares it with predefined threshold values. If abnormal temperature conditions or handling issues are detected, the LED indicators and buzzer alerts are activated immediately.

The system continuously repeats the monitoring cycle during transportation, ensuring vaccine safety and reliable delivery management.

3.3 FLOWCHART

Start



Initialize Sensors and System



Read Temperature, GPS, and



Process Data in Arduino Uno



Compare Data with Threshold Values

Yes → Continue monitoring and tracking

No → Activate LED and buzzer alert



Are Parameters Within Safe Range?



Repeat Monitoring Cycle



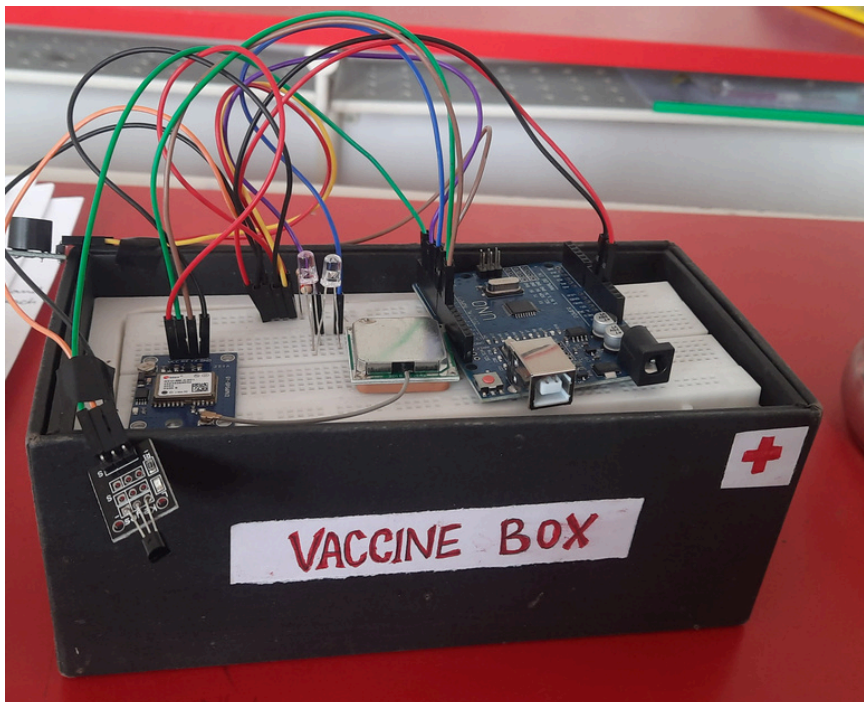
End

Project title: AUTOMATED VACCINE DELIVERY SYSTEM

3.4 PROTOTYPE DESCRIPTION

3.3.1 MATERIALS USED

- a) Arduino Uno (ATmega328P) – Main microcontroller
- b) DS18B20 Temperature Sensor – Monitors vaccine storage temperature
- c) NEO-6M GPS Module – Provides real-time location tracking
- d) LED Indicators – Status indication and warning signals
- e) Piezo Buzzer Module – Alert system for abnormal conditions
- f) 12V DC Power Supply – Main power source
- g) 5V Voltage Regulator – Provides stable voltage supply
- h) Breadboard – Circuit connections
- i) Jumper Wires – Electrical connection



CHAPTER 4

IMPLEMENTATION

- The implementation of the automated vaccine delivery system is based on an Arduino Uno microcontroller integrated with temperature monitoring, GPS tracking, and payload measurement modules. The Arduino continuously reads data from the DS18B20 temperature sensor, NEO-6M GPS module.

The collected data is processed and compared with predefined safe limits.

- Whenever the temperature exceeds the specified range or abnormal payload conditions are detected, the buzzer and LED indicators are activated immediately. The GPS module continuously provides real-time location updates of the vaccine delivery unit.

The system operates using a regulated 12V DC power supply with a 5V

- regulator for stable performance. The implementation ensures safe transportation, accurate monitoring, and efficient vaccine delivery management.

CHAPTER 5

CONCLUSION & FUTURE WORK

5.1 CONCLUSION

The automated vaccine delivery system provides an efficient and reliable solution for safe vaccine transportation by continuously monitoring important parameters such as temperature and delivery location. The system reduces dependence on manual monitoring and helps maintain proper vaccine storage conditions throughout transportation.

Real-time monitoring and automated alert mechanisms improve vaccine safety, reduce wastage, and support efficient healthcare logistics. This system is highly beneficial for hospitals, healthcare centers, vaccination campaigns, and medical supply transportation.

5.2 FUTURE SCOPE

In the future, the automated vaccine delivery system can be enhanced using IoT connectivity and cloud-based monitoring for remote access and centralized control. Artificial intelligence and predictive analytics can be integrated to improve delivery route optimization and fault detection.

Additional sensors such as humidity and shock sensors can be added to improve monitoring accuracy. Solar-powered operation and mobile application integration can further improve system efficiency, reliability, and sustainability in healthcare transportation systems.